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Tuning of oil well models with production data reconciliation

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ABSTRACT

The daily optimization of oil production systems relies on models to simulate phenomena of interest, such as fluid flow in wells and risers. Keeping models up to date is no easy task, due to the complex nature of the processes, and the need of human intervention to tune simulators. This work contributes by taking advantage of real-time measurements, to adjust process parameters that play a key role in steady-state simulators, e.g. the basic sediment and water and gas-oil ratio, and thereby optimize production for the prevailing conditions. The methodology consists of: (i) a strategy to identify steady-state of process variables (i.e., flows and pressures); (ii) an optimization formulation to adjust the key parameters such that the predicted total flows from the wells match the measured platform streams, while honoring pressure and flow measurements at critical points.

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1. Introduction

Several works from the literature propose methodologies for short-term steady-state optimization of oil production systems, which invariably rely on models that involve complex phenomena such as flows in wells and risers. According to the control-optimization hierarchy of oil production systems, these methodologies are often referred to as short-term and daily production optimization, since they concern the decisions that vary from days to weeks (Foss and Jensen, 2011). They constitute a valuable tool that helps engineers decide on platforms' best operating conditions, which depend on models for processes, wells, and reservoirs. Gunnerud and Foss (2010) present a methodology for short-term production optimization that accounts for routing decision and pressure balance, which leads to a mixed-integer linear programming (MILP) formulation. Hoffmann and Stanko (2017) developed an optimization formulation to maximize the oil production of platforms by defining optimal frequencies for electric submersible pumps. Grimstad et al. (2016) developed B-spline proxy models and a spatial branch-and-bound algorithm for direct solution of nonlinear production optimization problems, considering routing constraints as well as pressure and temperature relations.

The above works, in addition to others present in the literature, rely on the assumption that existing models represent the underlying processes' prevailing conditions. In practice, keeping models up to date is a challenging task, in part due to the complex and time-varying nature of the processes, but also because key parameters are only revised when a well is subject to a separation test, such as the gas-oil ratio (GOR) and basic sediment and water (BSW). As the well tests are performed only periodically, these key parameters may become outdated and not reflect the current conditions. The difficulties in model maintenance were motivational for Foss et al. (2018), who discussed the pros and cons of performing short-term production optimization as a dynamic or static problem. Their studies indicate that production optimization can be performed with static models, except in situations involving dynamic transients. This is the case of well shut-in and start-up operations as addressed by Hülse et al. (2020).

The works cited above acknowledge the importance of having models adjusted to reflect the prevailing conditions, which entails updating them with data from well tests and process measurements. Krishnamoorthy et al. (2018) address the case where transients are significant and the system seldom reaches a steady-state. Identification of the process parameters is achieved at the expense of working with dynamic models, which may be a rather complex task. Chachuat et al. (2009) review different approaches for making use of measurements to aid the production optimiza-

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tion problem. They argue in favor of the so called *modifier adaptation* in which the new measurements are used not only in tuning model parameters but also in modifying the optimization problem to more accurately define the feasible set. Jang et al. (1987) propose a "two-phase" method to identify model parameters using information from measurements and later employing the models to generate optimal operating conditions. This method is tested under uncertain models and sparse measurements of a chemical reactor by Chen and Joseph (1987). Câmara et al. (2016) analyze two industrial tools for real-time optimization applied to a process of oil distillation. Both tools attempt to detect the steady-state before making model updates. Bikmukhametov and Jäschke (2020) combine data-driven models and first principle models in a multiphase flow estimation problem.

This work follows the same strategy of Chachuat et al. (2009), Câmara et al. (2016) in the sense that a steady-state is identified to bring real-time information into the update of static models. The daily oil production optimization is a static problem as stated by Foss et al. (2018), which employs steady-state models. In this sense, we apply a l_1-trend filtering procedure (Kim et al., 2009) in addition to a steady-state detection method based on an adaptive polynomial (Rincón et al., 2015). The steady-state measurements are then used for data reconciliation in order to update key parameters of the static models, which enable static optimization more consistent with the operating conditions.

This model-updating strategy is also used here to address an important issue to oil production platforms: in general, it is not possible to measure the oil flow from each well separately. Oil-flow meters are usually only found after the production of multiple wells has already been mixed. This work presents a methodology to estimate individual production of wells that produce to a common separator, employing the available data on total production and pressures from key sensors and well simulation models. An optimization model is solved to adjust GOR and BSW parameters such that the predicted total flows from the wells match the measured platform streams while honoring pressure and flow measurements at critical points. Finally, the GOR and BSW parameters are updated in the well simulation models, which are then used to make daily production optimization more consistent with the prevailing operating conditions.

From a practical point of view, data reconciliation is essential for optimization procedures to drive production gains. To put another way, without correct parameter estimation (e.g., namely if the values of BSW and GOR are outdated), the optimization may lead to lower production. Decision support techniques in the oil industry, such as the one proposed here, do not render high percentage gains. However, even a numerically small gain of one per cent is significant, amounting to expressive monetary gains and driven without further investments, but rather obtained from optimized decisions.

1.1. Overview of methodology

The proposed methodology consists of the following steps:

1. Analysis of historical data from wells of interest:
   - Signal processing and filtering.
   - Identification of periods when the operation was in a steady state.
   - Estimating the value of the process variables that characterize the steady-states identified.
2. Estimation of well model parameters:
   - Input the steady-state operating conditions into the well simulation models.
   - Perform a sensitivity analysis of the simulation models with respect to the parameters of interest (e.g., GOR and BSW).
   - Obtain well production and pressure surrogates as a function of the parameters.
   - Solve an optimization problem to adjust the parameters so that the total production from wells matches system measurements.

1.2. Contributions

Everything considered, the paper's contribution is mostly technological, in the sense that the proposed methodology brings together existing tools from identification, modeling, and optimization to update key process parameters (namely GOR and BSW) that lead to improved operations. To a lesser degree, novelty can be attributed to the methodology itself in that it integrates distinct techniques into an effective model tuning tool for oil wells.

Concretely, we start with well-established tools of filtering and detection of signal trends by means of a l_1-trend filter, followed by an automatic procedure for steady-state identification based on adaptive polynomials. With well data measurements obtained by these procedures, we propose a method that relies on mathematical optimization and high-fidelity models for static well-model update (GOR and BSW). Those results combined can be used to render daily oil production optimization consistent with the current operating conditions, by means of data reconciliation.

1.3. Paper organization

The procedures for identification of the steady-state conditions and estimation of process variables are presented in Section 2. In Section 3, these variables are used to obtain sensitivity data for the parameters of interest from well simulation models, allowing the synthesis of an optimization formulation for updating the process parameters. The test procedures used to validate the methodology are presented in Section 4, in addition to the data from an actual production platform, which will serve as a case study. The methodology is then applied to the case study in Section 5, where numerical experiments are presented and discussed. Finally, the conclusions are presented in Section 6.

2. Steady-state identification

For the steady-state identification procedure, the key variables considered are the wellhead pressures and the lift-gas flows. This section presents the methodology for steady-state detection in a didactic way, using a well as an example along with the steps. Consider, for instance, the wellhead pressure given in Fig. 1, which is from an actual well operated by Petrobras. Unlike GOR and BSW values that are estimated only during well separation tests, the values referring to the controls of each well (wellhead pressure and lift-gas flows) and the combined flows of the platform (oil, gas and water) are periodically measured.

Notice that the signal is noisy, so the first step is to filter it. To that end, we propose an l_1 trend filter (Kim et al., 2009), in which every sample y_t at time $t \in [1, n]$ is substituted by a x_t which minimizes

$$f = \min_x \left(\frac{1}{2}\right) \sum_{t=1}^{n} (y_t - x_t)^2 + \lambda \sum_{t=2}^{n-1} |x_{t-1} - 2x_t + x_{t+1}|. \quad (1)$$

with $x = (x_1, \ldots, x_n)$ being the vector of variables. This formulation can be recast as a Quadratic Program (QP) and solved to optimality, for example, by means of a specific interior-point algorithm presented in Kim et al. (2009) or a commercial solver like Gurobi. The filtered signal is presented in Fig. 2 along with the original signal.

With the filtered signal in hand, we can proceed to the steady-state detection. Among the most employed steady-state detection

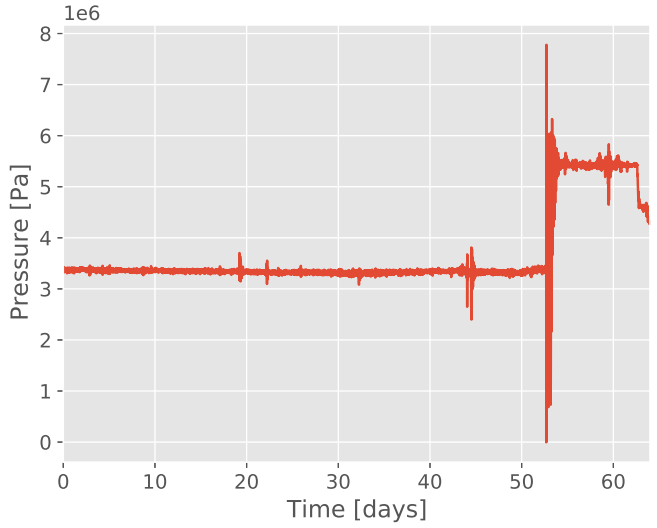


Fig. 1. Noisy wellhead pressure from an actual oil well operated by Petrobras.

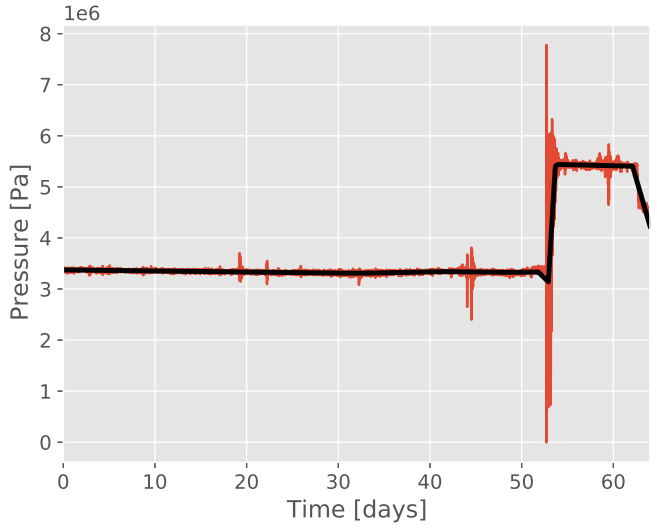


Fig. 2. Filtered wellhead pressure by means of a l_1 trend filter. The solid red line is the input signal and the black solid line is the filtered one.

methods, we can cite the F-like test and the adaptive polynomial (Rincón et al., 2015). We chose to employ the latter because, in benchmark tests presented in the literature, it performed better compared to other algorithms. In addition, when using the methodology as a tool in an oil and gas production facility, it becomes attractive for the user to have a steady-state detection method with few parameters to be configured.

The adaptive polynomial method first partitions the signal data into non overlapping windows, of constant size, and then fits a second-degree polynomial to each data window using linear regression (Tao et al., 2012). The linear coefficient of the polynomial is used as a metric to define whether the data characterizes a steady-state condition.

Now we formalize the procedure. For each window i of size N, let the sample times be $\{t_{j_i}, t_{j_i+1}, \ldots, t_{j_i+N-1}\}$, and let $p_i(t)$ be a polynomial that fits the signal through the samples of the interval:

$$p_i(t) = a_{i,0} + a_{i,1}(t - t_{j_i}) + a_{i,2}(t - t_{j_i})^2, \ \forall t \in \{t_{j_i}, \ldots, t_{j_i+N-1}\} \tag{2}$$

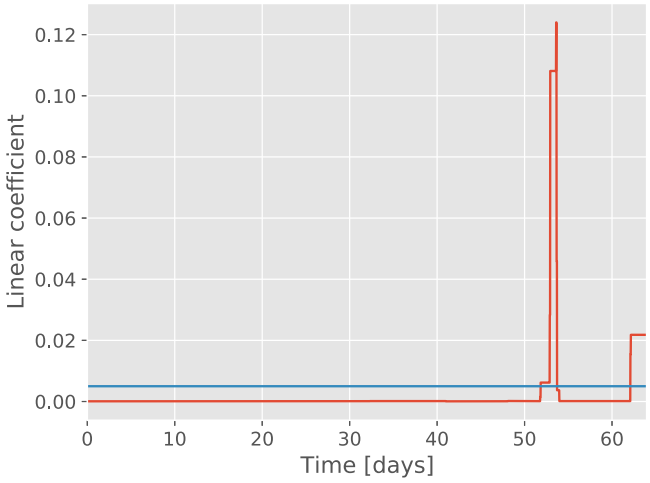


Fig. 3. Identified value of the linear coefficient $|a_{i,1}|$ of the fitted polynomial (absolute value, in red) and the threshold γ for steady-state identification (in blue). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

with the coefficients given by

$$\boldsymbol{\phi}_i = (a_{i,0}, a_{i,1}, a_{i,2})$$

and the variables by the vector

$$\mathbf{r}_i(t) = \left(1, (t - t_{j_i}), (t - t_{j_i})^2\right).$$

Notice the polynomial (2) can be written as:

$$p_i(t) = \mathbf{r}_i(t)^T \boldsymbol{\phi}_i \tag{3}$$

For any window, the coefficients $\boldsymbol{\phi}$ of the corresponding polynomial are obtained by solving (3) for all samples of the window, so

$$\begin{aligned}
x_{t_{j_i}} &= \mathbf{r}_i(t_{j_i})^T \boldsymbol{\phi}_i \\
x_{t_{j_i+1}} &= \mathbf{r}_i(t_{j_i+1})^T \boldsymbol{\phi}_i \\
&\vdots \\
x_{t_{j_i+N-2}} &= \mathbf{r}_i(t_{j_i+N-2})^T \boldsymbol{\phi}_i \\
x_{t_{j_i+N-1}} &= \mathbf{r}_i(t_{j_i+N-1})^T \boldsymbol{\phi}_i,
\end{aligned}$$

which can be written in matrix form as

$$\widetilde{\mathbf{x}}_i = R_i \boldsymbol{\phi}_i.$$

in which R_i is a constant matrix and $\widetilde{x}_i$ is a vector with the filtered variables.

Since the number of samples N is higher than 3 (the number of coefficients), the system has to be solved in the least-squares sense, for example with the use of the normal equation or QR decomposition (Golub and Loan, 1996). Either way, the solution will have the form

$$\widetilde{\boldsymbol{\phi}}_i = (R_i^T R_i)^{-1} R_i^T \widetilde{\mathbf{x}}_i.$$

Having obtained the coefficients of the polynomial, we consider the corresponding window i to be at steady-state if:

$$|a_{i,1}| < \gamma$$

with $\gamma > 0$ being a threshold parameter defined by the operator. The outcome of this step can be seen in Fig. 3, which plots the value of $|a_{i,1}|$ for the same signal shown before. The threshold γ is also shown. Finally, the steady-state decision is shown in Fig. 4, where the entered value is 0 if no stationary regime is identified (if the slope is greater than or equal to the defined threshold), or 1 if

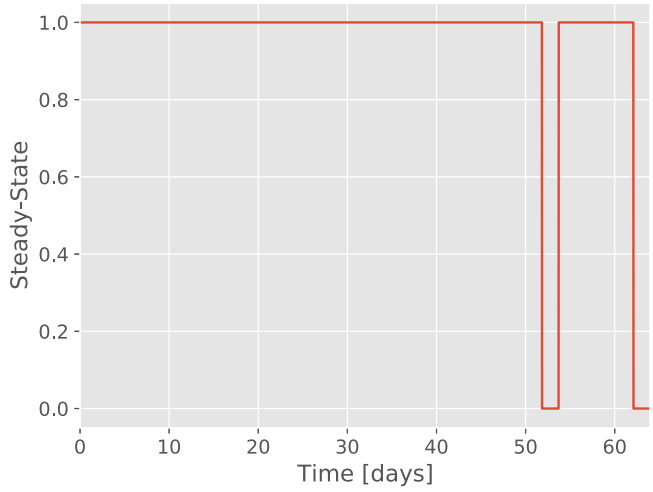


Fig. 4. Steady-state detection outcome: the value is 0 if no stationary regime is identified (if the slope is greater than or equal to the defined threshold), or 1 if a stationary regime is identified (the slope is less than the threshold).

a stationary regime is identified (the slope is less than the threshold). The steady-state identification considering (*i*) several variables for the same well and (*ii*) several wells on the same platform, can be treated in the same way. The simplest and most straightforward way is to consider an AND logic for the various slopes. Thus the steady-state condition happens when every variable/well is at "1" (i.e., the slope is below the threshold).

Though we have found instants for which the algorithm managed to find all wells in steady-state, this will be increasingly more difficult as more wells are routed to the same separator. In some cases, an automatic detection should include some kind of voting system, which should be investigated. Possibly, it suffices to consider when the wells at steady-state comprise 90% of the total flow.

3. Well model tuning with production reconciliation

This section presents a formulation for joint identification of key parameters of steady-state well models that accounts for field measurements of the operating conditions, such as pressures and input flows, while ensuring data reconciliation between the predicted and measured flows of the aggregated system production of oil, gas, and water. The well parameters of interest are GOR and BSW.

3.1. Production system modeling

The production reconciliation problem aims to approximate the predictions of the total production from the wells to the total measured production, in each separation train, when a steady-state operation is detected for the wells. For each well $n \in \mathcal{N}$ we search for the variables gor^n and bsw^n that are close to the values of GOR and BSW obtained from the latest well separation test. These values, hereafter called historical values, are denoted gor^n_{hist} and bsw^n_{hist}. Mathematically, this is obtained by minimizing the quadratic error:

$$f = \min_{(\mathbf{gor},\mathbf{bsw})} \sum_{n\in\mathcal{N}} \alpha^n_{\text{gor}} \cdot \left(gor^n - gor^n_{\text{hist}}\right)^2 + \sum_{n\in\mathcal{N}} \alpha^n_{\text{bsw}} \cdot \left(bsw^n - bsw^n_{\text{hist}}\right)^2 \tag{4}$$

where $\mathcal{N}$ is the set of wells, and α_{gor} and α_{bsw} are positive weights for the errors. Weights are used to enable degrees of freedom for the operator. For example, from his/her knowledge of the process plant the engineer can infer that the GOR values are more reliable than the BSW values, or vice versa. Thus, he/she can give greater weight to the variables of greater confidence.

3.1.1. Production from wells

Although complex systems could be considered, possibly with subsea manifolds and electric submersible pumps, the developments hereafter focus on platforms with only satellite wells to emphasize the methodology's key elements for well parameter tuning with data reconciliation. The platforms of interest operate with gas-lifted wells, which have a liquid production induced by the wellhead pressure whp and lift-gas injection rate q_{gl}. The following equations model the individual production for each well $n \in \mathcal{N}$:

$$q^{n,t}_{\text{w}} + q^{n,t}_{\text{o}} = \widehat{q}^n_{\text{l}}(q^{n,t}_{\text{gl}}, whp^{n,t}, bsw^n, gor^n), \tag{5a}$$

$$q^{n,t}_{\text{g}} = gor^n \cdot q^{n,t}_{\text{o}}, \tag{5b}$$

$$q^{n,t}_{\text{w}} \cdot (1 - bsw^n) = bsw^n \cdot q^{n,t}_{\text{o}}, \tag{5c}$$

for a given time $t \in \mathcal{T}$, where $\widehat{q}^n_{\text{l}}$ is the liquid production function and $q^{n,t}_o$, $q^{n,t}_g$, $q^{n,t}_w$ being the phase flow for oil, gas and water, respectively. Constraint (5a) defines the liquid production in terms of the oil and water produced by the reservoir[1], and (5b) and (5c) give the phase flows for produced gas and water. Notice that $bsw^n = q^n_w/q^n_l$.

3.1.2. Production trains

Each production train has its own separator, and the total production of oil, gas, and water is known. Let $\mathcal{C}$ denote the set of all such trains, and $\mathcal{N}(c)$ be the set of wells that have their production routed to a train c. Notice that $\cup_{c\in\mathcal{C}}\mathcal{N}(c) = \mathcal{N}$ and $\mathcal{N}(c) \cap \mathcal{N}(c') = \emptyset$. The production of each train $c \in \mathcal{C}$ corresponds to the sum of the production of each well routed the train, namely $n \in \mathcal{N}(c)$. Therefore, for a given time ($t \in \mathcal{T}$) where the system is in steady-state:

$$q^{c,t}_{\text{oil-prod}} = \sum_{n\in\mathcal{N}(c)} q^{n,t}_{\text{o}}, \tag{6a}$$

$$q^{c,t}_{\text{gas-prod}} = \sum_{n\in\mathcal{N}(c)} q^{n,t}_{\text{g}}, \tag{6b}$$

$$q^{c,t}_{\text{water-prod}} = \sum_{n\in\mathcal{N}(c)} q^{n,t}_{\text{w}}. \tag{6c}$$

Similarly the usage of lift-gas of a production train is the sum of the lift-gas rates of individual wells of the train:

$$q^{c,t}_{\text{gas-lift}} = \sum_{n\in\mathcal{N}(c)} q^{n,t}_{\text{gl}}. \tag{7}$$

Fig. 5 illustrates the schematics of an offshore platform that produces from gas-lifted wells, which can have their production routed to distinct production trains.

Typically, the $q^t_{\text{gas-prod}}$ is not directly measured in a platform. Instead the produced gas is calculated considering the gas exported, the gas burned in the flare and the consumption in the turbine, as shown in the following equation:

$$q^t_{\text{gas-prod}} = q^t_{\text{gas-exp}} + q^t_{\text{flare}} - q^t_{\text{turbine}}. \tag{8}$$

Notice that this formulation is possible because we are not considering gas-injection wells.

[1] Unlike other works, the well production is modeled by means of liquid flow $\widehat{q}_{\text{l}}(\cdot)$ instead of oil rate, to not constrain degrees of freedom.

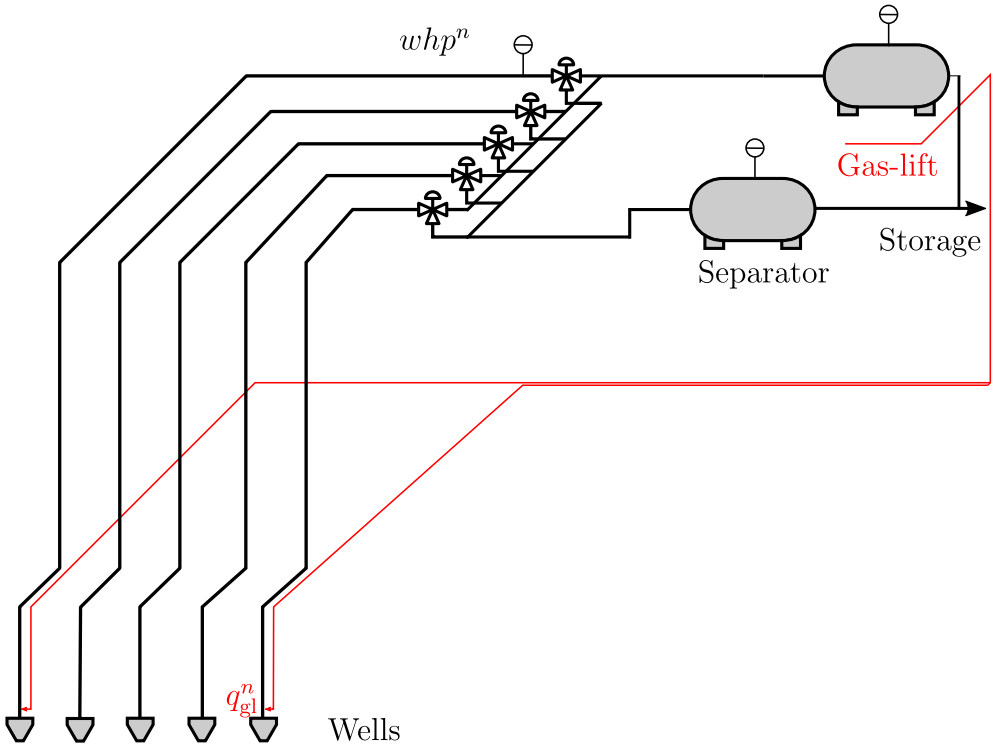


Fig. 5. Platform scheme. Wells (shown at the bottom) produce a mixture of oil, water and gas that is lifted through pipelines and can be routed to either separator. Individual well measurements for these flows are not available. whp^n is the wellhead pressure of well n. Measurements $q_x^{c,t}$, with $x \in$ {oil-prod, water-prod, gas-prod} are taken at the separator. q^n_{gl} is the flow of lift-gas injected into well n.

3.1.3. Platform

The total production of the platform is given by the sum of the production from each train of production ($c \in \mathcal{C}$). The same for consumption of lift-gas. The following constraints state that for each time $t \in \mathcal{T}$ in which the production of the platform is stationary:

$$q^t_{\text{oil-prod}} = \sum_{c\in\mathcal{C}} q^{c,t}_{\text{oil-prod}}, \tag{9a}$$

$$q^t_{\text{gas-prod}} = \sum_{c\in\mathcal{C}} q^{c,t}_{\text{gas-prod}}, \tag{9b}$$

$$q^t_{\text{water-prod}} = \sum_{c\in\mathcal{C}} q^{c,t}_{\text{water-prod}}, \tag{9c}$$

$$q^t_{\text{gas-lift}} = \sum_{c\in\mathcal{C}} q^{c,t}_{\text{gas-lift}} \tag{9d}$$

The remaining constraints are also related to platform limitations (*e.g.*, capacity of compression of gas and capacity of liquid and water treatment):

$$q^t_{\text{gas-prod}} + q^t_{\text{gas-lift}} - q^t_{\text{flare}} \le q^t_{\text{gtc}}, \tag{10a}$$

$$q^t_{\text{gas-exp}} \le q^{\max}_{\text{gas-exp}}, \tag{10b}$$

$$q^{\min}_{\text{flare}} \le q^t_{\text{flare}} \le q^{\max}_{\text{flare}}, \tag{10c}$$

$$q^{c,t}_{\text{oil-prod}} + q^{c,t}_{\text{water-prod}} \le q^{c,\max}_{\text{l}}, \tag{10d}$$

$$q^{c,t}_{\text{water-prod}} \le q^{c,\max}_{\text{w}}, \tag{10e}$$

$$q^{c,t}_{\text{oil-prod}} \le q^{c}_{\text{o,max}}, \tag{10f}$$

$$q^{c,t}_{\text{gas-prod}} \le q^{c}_{\text{g,max}}. \tag{10g}$$

3.1.4. Operating conditions

The wellhead pressure and gas-lift flow rate were fixed at the values identified as part of the steady-state. So for each $t \in \mathcal{T}$ the constraints

$$whp^{n,t} = whp^{n,t}_{\text{st}}, \tag{11a}$$

$$q^{n,t}_{\text{gl}} = q^{n,t}_{\text{gl,st}}, \tag{11b}$$

refer to the well operating conditions ($\forall n \in \mathcal{N}(c),\ \forall c \in \mathcal{C}$). The following constraints establish the ranges within which the identified GOR and BSW are allowed to assume their values:

$$gor^n_{\min} \le gor^n \le gor^n_{\max}, \tag{12a}$$

$$bsw^n_{\min} \le bsw^n \le bsw^n_{\max}. \tag{12b}$$

Data reconciliation is achieved by ensuring that production from each train $c \in \mathcal{C}$, as calculated in Eq. (6), matches the corresponding production measured at steady-state conditions. Our model allows a small tolerance ϵ in relation to the value measured, leading to

$$q^{c,t,\text{st}}_{\text{oil-prod}} \cdot (1-\epsilon_{\text{oil}}) \le q^{c,t}_{\text{oil-prod}} \le q^{c,t,\text{st}}_{\text{oil-prod}} \cdot (1+\epsilon_{\text{oil}}), \tag{13a}$$

$$q^{c,t,\text{st}}_{\text{gas-prod}} \cdot (1-\epsilon_{\text{gas}}) \le q^{c,t}_{\text{gas-prod}} \le q^{c,t,\text{st}}_{\text{gas-prod}} \cdot (1+\epsilon_{\text{gas}}), \tag{13b}$$

$$q^{c,t,\text{st}}_{\text{water-prod}} \cdot (1-\epsilon_{\text{water}}) \le q^{c,t}_{\text{water-prod}} \le q^{c,t,\text{st}}_{\text{water-prod}} \cdot (1+\epsilon_{\text{water}}), \tag{13c}$$

for each production train $c \in \mathcal{C}$ and steady-state condition $t \in \mathcal{T}$. Notice that $\boldsymbol{\epsilon} = \{\epsilon_{\text{oil}}, \epsilon_{\text{gas}}, \epsilon_{\text{water}}\}$ define the priority given by the operator to each of the measurements of the platform. The operator can define a tighter $\boldsymbol{\epsilon}$ for the oil, while giving water and gas reconciliation more freedom, as he/she understands that the latter are more inaccurate. Thus, while tight bounds for $\boldsymbol{\epsilon}$ are imposed in the form of constraints, the cost function causes these bounds to be achieved in a way that the operating point is less distant from the historical values (i.e., the latest separation tests).

Initial tests also considered the reconciliation variables in the cost function (i.e., $(y_{\text{measured}} - y)^2$). However, these tests revealed that this approach was unsatisfactory. Since, for instance, in real measurements of a platform, there is more precision in measurements of oil than gas, and especially water.

Also, the mix of several variables — involving oil, gas, water and historical values — in the cost function, led to optimization results that did not yield good reconciliation. For example, the results for some instances tried to satisfy historical values, but not adequately reconciling oil, or even prioritizing water over oil, which is not a good option due to the measurements accuracy.

3.2. Well tuning optimization

3.2.1. Conceptual formulation

The problem of tuning well parameters with production system data reconciliation can be formalized as follows:

$$f = \min_{(\mathbf{gor},\mathbf{bsw})} \sum_{n\in\mathcal{N}} \alpha^n_{\text{gor}} \cdot \left(gor^n - gor^n_{\text{hist}}\right)^2 + \sum_{n\in\mathcal{N}} \alpha^n_{\text{bsw}} \cdot \left(bsw^n - bsw^n_{\text{hist}}\right)^2 \tag{14a}$$

$$\text{s.t.: Eqs. (5)} -\!- \text{(13)} \tag{14b}$$

where $\mathbf{gor} = (gor^n : n \in \mathcal{N})$ and $\mathbf{bsw} = (bsw^n : n \in \mathcal{N})$ are vectors with decision variables. The well tuning parameter problem is *conceptual* in the sense that the well liquid production function $\widehat{q}_{\text{l}}(\cdot)$ is not explicitly given in a form that can be directly optimized. The problem is also nonlinear due to the bilinear terms that result from the product of flow rates with GOR and BSW in Eqs. (5b) and (5c).

The liquid flow function $\widehat{q}_{\text{l}}(\cdot)$ is typically available in a simulator which models the physical structure of the well, the well-reservoir interface, and the artificial lifting method such as gas-lift. By solving complex algebraic and partial differential equations, the simulator can represent these dynamics of flow (*e.g.*, OLGA (Schlumberger, 2012)) or the steady conditions (*e.g.*, PIPESIM (Schlumberger, 2009) and MARLIM (Seman et al., 2020)), with the latter being the focus of this work. Derivative-free optimization could then be applied to solve the well tuning problem by replacing $\widehat{q}_{\text{l}}(\cdot)$ with a suitable simulation model. This approach imposes difficulties regarding the computational cost of simulating multiple wells online, the large number of simulator licenses required, and the potential lack of efficiency of derivative-free methods (Merlini Giuliani and Camponogara, 2015; Seman et al., 2020).

An alternative that has been successful in short-term production optimization is the piecewise-linear approximation of the production function $\widehat{q}_{\text{l}}(\cdot)$. It consists in sampling the simulation model of a well and performing a sensitivity analysis on the parameters of interest (gor^n, bsw^n), while holding the operating conditions $(whp^{n,t}_{\text{st}}, q^{n,t}_{\text{gl,st}})$ fixed (c.f., Eq. (11)), and then producing the piecewise-linear model. As a piecewise-linear approximation introduces binary variables to the model (Vielma et al., 2010), it would become a mixed-integer problem (Silva and Camponogara, 2014). Thus, the resulting problem would be a MINLP, since in addition to the binary variables introduced by the piecewise-linearization, there would be also the product of variables in Eqs. (5b) and (5c). A MILP could be recovered by modeling the bilinear terms $gor^n \cdot q^{n,t}_{\text{o}}$, $bsw^n \cdot q^{n,t}_{\text{w}}$, and $(1 - bsw^n) \cdot q^{n,t}_{\text{o}}$ with two-dimensional piecewise-linear functions. The initial difficulty at solving the MINLP and approximation MILP with state-of-the-art solvers, such as Gurobi (Gurobi Optimization, 2020) and CPLEX (IBM ILOG CPLEX, 2009), motivated an alternative approach detailed below.

3.2.2. Simplified well model

Most of the constraints of the model above are linear, with exception of (5a), (5b) and (5c). The hardest of these is the liquid production (5a), which may be the result of complex computer simulations. After experimenting with a piecewise-linear approximation of the liquid flow, we settled with a single quadratic of the form:

$$\widehat{q}^n_{\text{l}}(q^{n,t}_{\text{gl}}, whp^{n,t}, bsw^n, gor^n) \approx a^{n,t}_0 + a^{n,t}_1 gor^n + a^{n,t}_2 (gor^n)^2 + a^{n,t}_3 bsw^n + a^{n,t}_4 (bsw^n)^2 + a^{n,t}_5 bsw^n gor^n, \tag{15}$$

where the coefficients $a^{n,t}_0, \ldots, a^{n,t}_5$ are identified by linear least-squares for each operating condition t. Given the nature of the liquid production as a function of *gor* and *bsw*, with *whp* and q_{gl} fixed, the proposed regression is able to achieve high fitting scores, as illustrated in Fig. 6.

3.2.3. Practical optimization formulation

For the tuning of GOR and BSW parameters of all wells, we solve the problem of minimizing objective (4) subject to constraints (5)–(15). Formally:

$$\min_{(\mathbf{gor},\mathbf{bsw})} f = \sum_{n\in\mathcal{N}} \alpha^n_{\text{gor}} \cdot \left(gor^n - gor^n_{\text{hist}}\right)^2 + \sum_{n\in\mathcal{N}} \alpha^n_{\text{bsw}} \cdot \left(bsw^n - bsw^n_{\text{hist}}\right)^2 \tag{16a}$$

$$\text{s.t. Eqs.(5)} -\!- \text{(13)} \tag{16b}$$

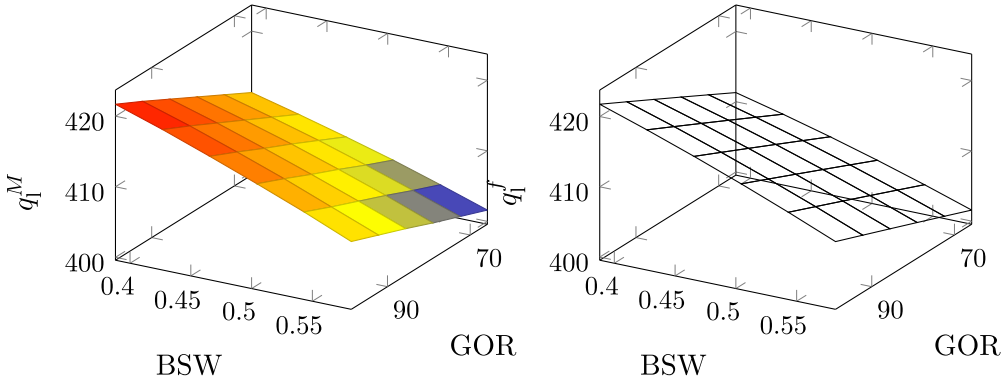
$$\text{Eq. (15)} \tag{16c}$$

which defines the problem of simultaneous tuning well models with system production reconciliation. Problem (16) is a special case of NLP with all nonlinear terms being bilinear. Nonconvex nonlinear terms appear only in the constraints (5b), (5c), and (15), whereas the nonlinear terms of the objective are convex. Global solvers can be used to solve (16), such as a nonconvex Gurobi, Baron (Sahinidis, 1996), and Couenne (Belotti et al., 2009). In the face of computational difficulties (Floudas, 1995), local solvers like IPOPT (Wächter and Biegler, 2006) and decomposition strategies may be considered. The latter would result in the solution of a relaxation obtained by replacing bilinear terms with McCormick envelopes (McCormick, 1976) to obtain a bound and initial guess, which is then used by a local NLP solver.

The variables to be defined by the operator in the practical optimization problem (16) are $\boldsymbol{\alpha}$ and $\boldsymbol{\epsilon}$, which concern, respectively, to the weights given to deviations from the historical values of GOR and BSW; and to the bounds of the reconciliation values of interest (i.e., oil, gas and water).

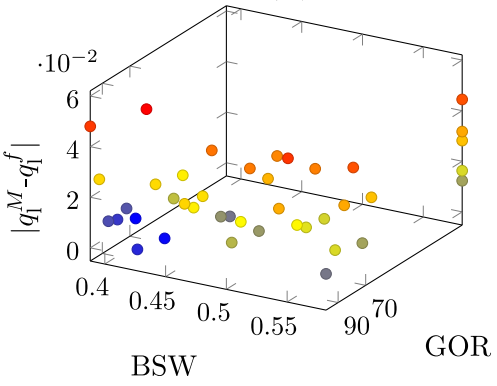
Typically, operators using methodologies such as those presented in this paper are highly specialized, and take into account specific knowledge about their plants. Thus, although poorly configured parameters can negatively influence the solution, we understand that typically the opposite happens. In other words, the operator's feeling and experience with the plant in question can make it possible for him/her to fine-tune the parameters in the most appropriate way possible.

With updated GOR and BSW values, the operators can make informed decisions on the platform's future production. To assess the quality of such decisions, the next Section presents a comparison



(a) Liquid production given by the MARLIM simulator, from Petrobras.

(b) Liquid production given by the quadratic model (15) identified with regression.



(c) Absolute difference between the liquid production given by MARLIM and the quadratic model.

Fig. 6. The liquid production q_l of one well, in m^3/d, for a given steady-state condition.

between the production driven by updated and historical parameters.

Notice that the problems discussed in this work are formally defined in mathematical programming. The problem classes of concern in this paper are: (*i*) NLP is a class of optimization problems with continuous variables, which have nonlinear functions in the objective and/or the constraints; they can be solved by efficient algorithms such as the interior-point method (Wächter and Biegler, 2006) to reach a local optimum, but which do not have a guarantee of a global optimum. (*ii*) MINLP is similar to NLP, with the addition of integer variables; although the global optimum of MINLP and NLP can possibly be found by algorithms like spatial branch-and-bound, combined with convex underestimators such as McCormick envelopes (McCormick, 1976), the computational cost can be unaffordable (Floudas, 1995). (*iii*) MILP refers to problems with continuous and discrete variables, but only linear constraints and objective function, which offer more structure for solution methods; the global optimum of a MILP model can be obtained by efficient algorithms like branch-and-bound and branch-and-cut, available in most state-of-the-art solvers (Gurobi Optimization, 2020).

In this paper we make use of two solvers in order to solve the problems of interest: (*i*) IPOPT (Wächter and Biegler, 2006) to deal with NLPs by means of interior-point methods; (*ii*) Gurobi (Gurobi Optimization, 2020) to deal with MILPs, which employs state-of-the-art methods including branch-and-bound and branch-and-cut. Notice that MINLPs are not directly solved in this paper, since techniques are used to transform them into the aforementioned problems.

4. Test procedure

This section presents the procedure designed to validate the well-tuning methodology, with data reconciliation, in an oil production platform. We first present the data (production system) and the setup that are used in the tests. Next, we explain a projection procedure used to carry out the validation of the results in comparison with the ideal values.

4.1. Production system

The oil platform of the case study consists of a single group of wells ($\mathcal{C} = \{c\}$ is a unit set) producing to a common train of production. Several scenarios were created, with the number of wells in c varying from 4 to 10.

The wells of each scenario are a subset of 10 oil wells that are producing in an actual platform operated by Petrobras in the Campos Basin. The constructive characteristics of the wells were kept unaltered, while the following features were varied randomly: (*i*) benchmark values for GOR and BSW at a pseudo present time; (*ii*) steady-state conditions with gas-lift rate q_{gl} and wellhead pressure *whp*.

The wells' detailed physical representation, the interface with reservoir, and the produced fluids are modeled in the MARLIM simulator from Petrobras. The simulator is routinely used for sensitivity and flow assurance analyses of naturally flowing wells and wells equipped with continuous gas lift. Once the well configuration is defined, the simulator computes one of the following three variables of interest given the other two: equilibrium flow, down-

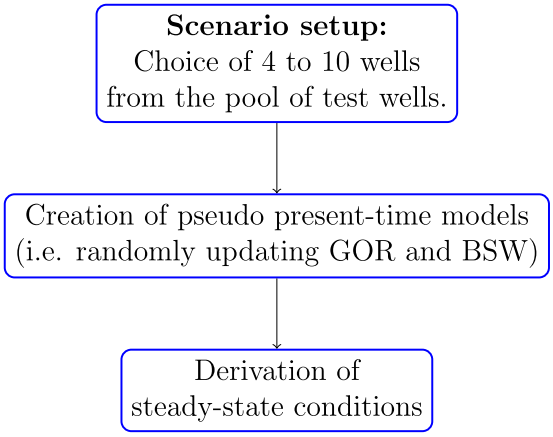


Fig. 7. Steps for the synthesis of a scenario.

stream pressure, or upstream pressure. The calculations of profiles are performed for flow pressure (using correlations) and temperature along the pipelines. The operating conditions of each subsurface valve are also computed, including gas flows (Seman et al., 2020).

We considered only one steady-state for each scenario, meaning $\mathcal{T} = \{t\}$ is a unit set containing only the present time. Once the steady-state values for gas-lift and wellhead pressure were defined, the performance relations of Eq. (15) were identified by linear least-squares employing data obtained from the Marlim simulator.

4.2. Experimental setup

The experiments were implemented in Python 3.6. They were run in a Ubuntu environment, and the computer had two Intel Core Xeon E5-2630 v4 Processors (2.20 GHz), adding up to 20 cores of 2 threads and 64 GB of RAM.

The proposed methodology was tested with different scenarios, each of which was created using the steps illustrated in Fig. 7, which are summarized as:

- First, a random set of 4 to 10 wells is chosen among the pool of test wells. Notice that all the well models from the pool originally have their GOR and BSW values defined as the values measured on the last separator test, which we will consider as historical values.
- The wells are then updated to a pseudo present-time GOR and BSW, defined as benchmark values, which are set at random values in a neighborhood of the historical values.
- The steady-state conditions (with the wellhead pressures and lift-gas rates of Eq. (11)) are also chosen randomly within a predetermined range and the pseudo present-time models are simulated in Marlim, rendering the production of oil, gas and water at the present time.

These production rates are added over all operating wells, rendering the overall system production for the steady-state condition as follows

$$q^{c,t,\mathrm{st}}_{\mathrm{oil-prod}} = \sum_{n\in\mathcal{N}(c)} q^{n,t,\mathrm{st}}_{\mathrm{o}},$$

$$q^{c,t,\mathrm{st}}_{\mathrm{gas-prod}} = \sum_{n\in\mathcal{N}(c)} q^{n,t,\mathrm{st}}_{\mathrm{g}},$$

$$q^{c,t,\mathrm{st}}_{\mathrm{water-prod}} = \sum_{n\in\mathcal{N}(c)} q^{n,t,\mathrm{st}}_{\mathrm{w}}.$$

This defines the variables $q^{c,t,\mathrm{st}}_{\mathrm{oil-prod}}$, $q^{c,t,\mathrm{st}}_{\mathrm{gas-prod}}$ and $q^{c,t,\mathrm{st}}_{\mathrm{water-prod}}$ of Eq. (13) in terms of the individual flows of the wells. Despite the randomly disturbed GOR and BSW values involved in the testing processes, they are derived to be consistent with the separation tests values in the pool of test wells (i.e., within a neighborhood of the historical values). Thus, the models involved still keep magnitudes and characteristics consistent with those of a real platform.

Notice that the models of wells with benchmark values are considered the plant and would effectively represent the production platform's behavior in a certain moment of time. Although typically unknown, except during the separation test, the benchmark values are used here as comparison parameters.

4.3. Projection procedure

As a way to validate the results achieved by the reconciliation methodology, we present next a methodology for projecting the results and their respective comparison with an ideal model (benchmark values). Thus, first we present the classic problem of maximizing oil in a production system (given previously known values of GOR ans BSW), to later derive the methodology for projecting the results.

4.3.1. Oil production maximization

The presented reconciliation methodology renders adjusted values for GOR and BSW of each well (gor^n_{adj}, bsw^n_{adj}). These values can be used in the daily production optimization of the platform. This is achieved using a general-purpose model in use by Petrobras and not limited to the present work. It will be formalized as follows. The objective is to maximize total oil production:

$$\max_{(\mathbf{q}_{\mathrm{gl}},\mathbf{whp})} \sum_{c\in\mathcal{C},t\in\mathcal{T}} q^{c,t}_{\mathrm{oil-prod}}. \tag{17}$$

The problem inherits most of the constraints from the previous one, namely (5)–(10). Constraints (11) are not included, since the well-head pressures and lift-gas rates become decision variables. Constraints (12) are replaced, as the corresponding variables are known:

$$gor^n_{\mathrm{min}} = gor^n = gor^n_{\mathrm{max}} = gor^n_{\mathrm{adj}}$$

$$bsw^n_{\mathrm{min}} = bsw^n = bsw^n_{\mathrm{max}} = bsw^n_{\mathrm{adj}}$$

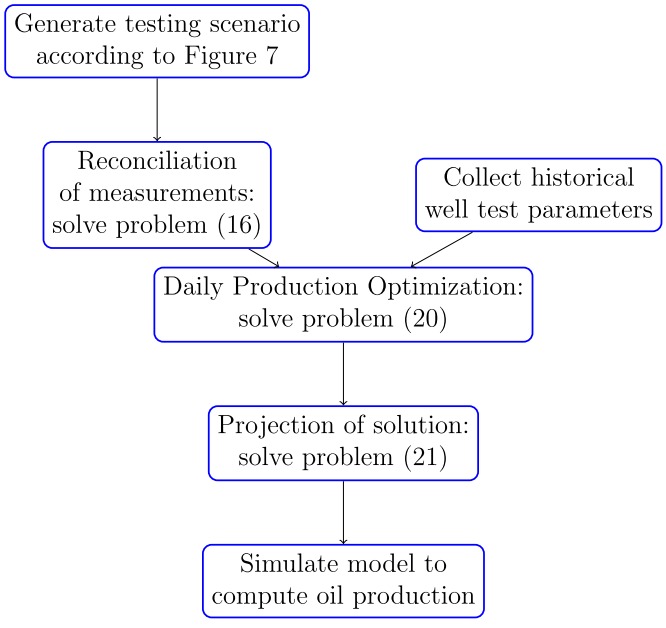


Fig. 8. Steps of the numerical experiments.

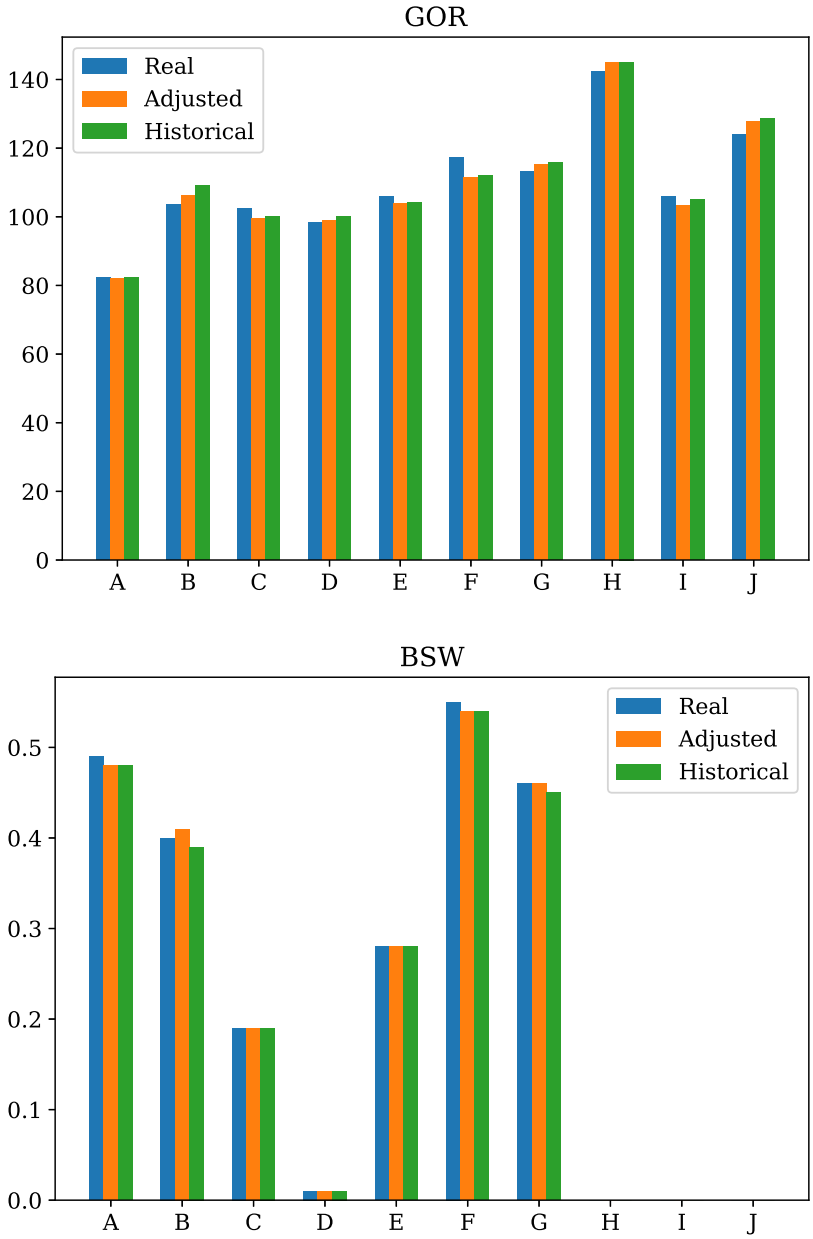


Fig. 9. GOR and BSW for a set of wells: comparing the adjusted results of the proposed methodology with the values of the historical well-test values and the actual (error-free) parameters. Notice that the values of BSW for wells H, I and J are zero or approximately zero.

Constraints (13) are also not used, since the actual flow values depend on the optimal values. Constraint (15) is also removed.

The identified values of GOR and BSW for each well are used to update the MARLIM simulation models, obtain production flows from sensitivity analyses, and then yield approximations for liquid production $\widehat{q}_{\mathrm{l}}^{n}(q_{\mathrm{gl}}^{n}, whp^{n},\ bsw_{\mathrm{adj}}^{n}, gor_{\mathrm{adj}}^{n})$ as a function of the lift-gas injection and well-head pressure, that is, $(q_{\mathrm{gl}}^{n}, whp^{n})$. The simplified model for liquid production $\widehat{q}_{\mathrm{l}}^{n}(q_{\mathrm{gl}}^{n}, whp^{n})$ will not be a quadratic model (as was in (15)), but a piecewise-linear curve.

By employing piecewise-linearization we cast a problem which off-the-shelf solvers can be used to provide a global optimum. Since it does not have bilinear terms such as the reconciliation problem, the use of $(\mathbf{q}_{\mathrm{gl}}, \mathbf{whp})$ points from the simulator in the construction of the approximation curves allows the search for the global optimum using efficient methods like branch-and-bound.

Let $\mathcal{K}_{\mathrm{gl}} = \{1, \ldots, k_{\mathrm{gl}}\}$ and $\mathcal{J}_{\mathrm{whp}} = \{1, \ldots, j_{\mathrm{whp}}\}$ be set of indices for lift-gas injection and well-head pressure. Let also $\mathcal{Q}_{\mathrm{gl}}^{n} = \{q_{\mathrm{gl}}^{n,k} : k \in \mathcal{K}_{\mathrm{gl}}\}$ and $\mathcal{W}_{\mathrm{whp}}^{n} = \{whp^{n,j} : j \in \mathcal{J}_{\mathrm{whp}}\}$ be the respective breakpoint values for well n, and

$$\mathcal{Q}_{\mathrm{liq}}^{n} = \left\{ q_{\mathrm{l}}^{n,k,j} : q_{\mathrm{l}}^{n,k,j} = \widehat{q}_{\mathrm{l}}^{n}(q_{\mathrm{gl}}^{n,k}, whp^{n,j}, bsw_{\mathrm{adj}}^{n}, gor_{\mathrm{adj}}^{n}), \quad k \in \mathcal{K}_{\mathrm{gl}}, j \in \mathcal{J}_{\mathrm{whp}} \right\}$$

be the liquid flow rates for the well at varying operating conditions (defined by breakpoints), which are obtained by using the MARLIM simulator with the adjusted parameters for GOR and BSW.

Generically, let $f : \mathcal{X} \to \mathbb{R}$ be a function that maps from the domain $\mathcal{X} \subset \mathbb{R}$ to the real numbers. Thus f is a function with an one-dimensional domain. Let $\mathcal{X}' \subset \mathcal{X}$ be an ordered set of points x for which $f(x)$ is known, e.g. $\mathcal{X}' = \{x^1, x^2, \ldots, x^{|\mathcal{X}'|}\}$. The breakpoints are the elements $x \in \mathcal{X}'$, for which the function values $f(x)$ are

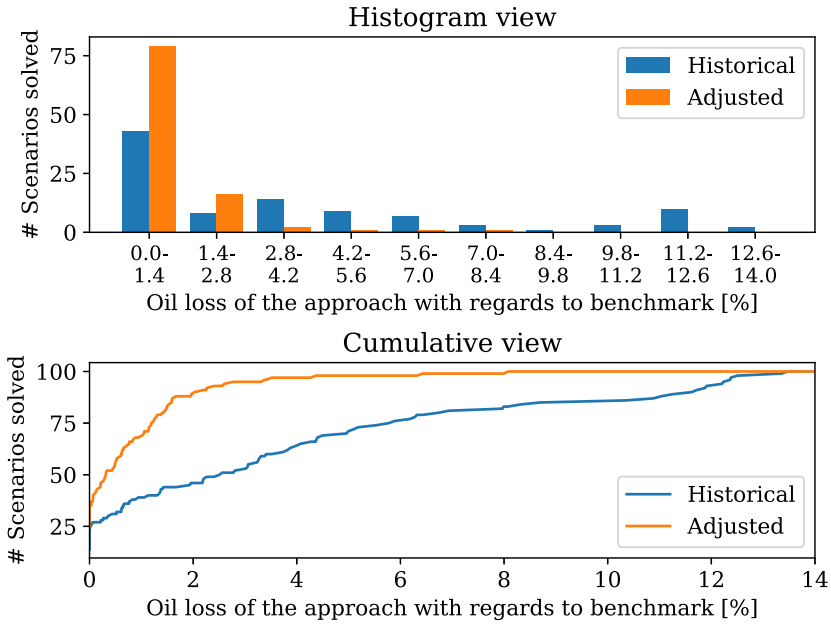


Fig. 10. Number of solutions of each approach, given the loss with respect to the benchmark. Histogram and cumulative view. Case with disturbance of 5%.

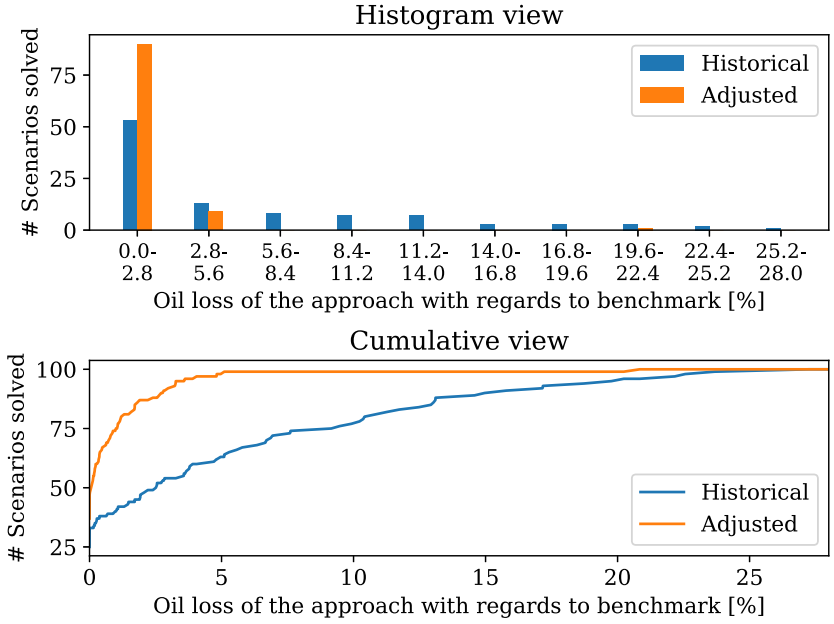


Fig. 11. Number of solutions of each approach, given the loss with respect to the benchmark. Histogram and cumulative view. Case with disturbance of 10%.

known and will be use to yield an approximation $\tilde{f}$ to f for the entire domain.

Formally, for each well $n \in \mathcal{N}$, the piecewise-linear approximation is given by

$$q_{\text{gl}}^{n} = \sum_{k \in \mathcal{K}_{\text{gl}}} \sum_{j \in \mathcal{J}_{\text{whp}}} \theta_{k,j}^{n} q_{\text{gl}}^{n,k} \tag{18a}$$

$$whp^{n} = \sum_{k \in \mathcal{K}_{\text{gl}}} \sum_{j \in \mathcal{J}_{\text{whp}}} \theta_{k,j}^{n} whp^{n,j} \tag{18b}$$

$$q_{\text{l}}^{n} = \sum_{k \in \mathcal{K}_{\text{gl}}} \sum_{j \in \mathcal{J}_{\text{whp}}} \theta_{k,j}^{n} q_{\text{l}}^{n,k,j} \tag{18c}$$

$$1 = \sum_{k \in \mathcal{K}_{\text{gl}}} \sum_{j \in \mathcal{J}_{\text{whp}}} \theta_{k,j}^{n} \tag{18d}$$

$$\theta_{k,j}^{n} \geq 0, \ k \in \mathcal{K}_{\text{gl}}, \ j \in \mathcal{J}_{\text{whp}} \tag{18e}$$

and SOS2 type constraints are necessary to ensure a piecewise linear approximation as follows

$$\eta_{k}^{n} = \sum_{j \in \mathcal{J}_{\text{whp}}} \theta_{k,j}^{n}, \ k \in \mathcal{K}_{\text{gl}} \tag{19a}$$

$$\eta_{j}^{n} = \sum_{k \in \mathcal{K}_{\text{gl}}} \theta_{k,j}^{n}, \ j \in \mathcal{J}_{\text{whp}} \tag{19b}$$

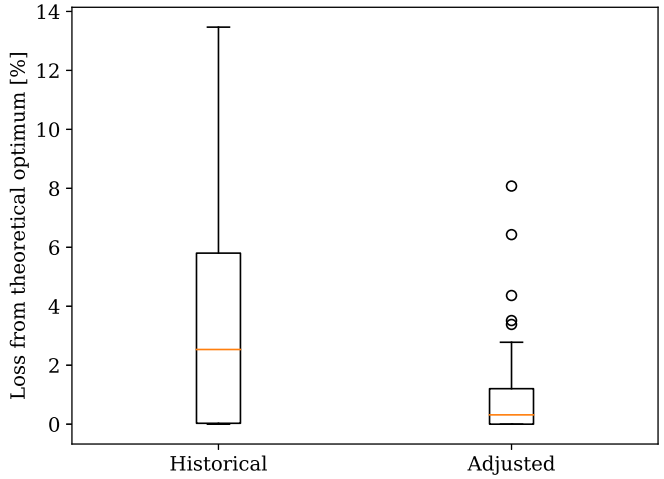


Fig. 12. Distribution of oil loss with respect to theoretical optimum, for both approaches. Case with disturbance of 5%.

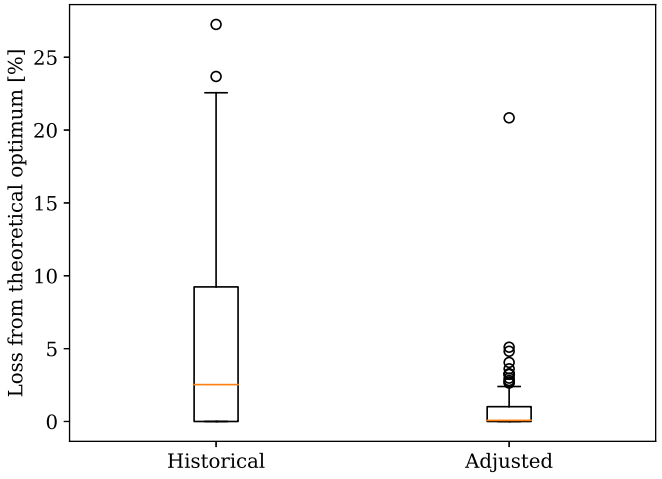


Fig. 13. Distribution of oil loss with respect to theoretical optimum, for both approaches. Case with disturbance of 10%.

$$\left\{\eta_k^n\right\}_{k\in\mathcal{K}_{\text{gl}}} \text{ is SOS2} \tag{19c}$$

$$\left\{\eta_j^n\right\}_{j\in\mathcal{J}_{\text{whp}}} \text{ is SOS2} \tag{19d}$$

SOS2 stands for specially ordered sets of type 2 (Beale and Tomlin, 1970), which means that at most two elements of the ordered set can assume nonzero values, in which case they must be consecutive in the order.

Thus, the problem of maximizing oil production is given by

$$\max_{(\mathbf{q}_{\text{gl}},\mathbf{whp})} \ (17) \tag{20a}$$

$$\text{s.t. } (5)\text{–}(10) \tag{20b}$$

$$(18),\ (19) \tag{20c}$$

which is a MILP. The solution of this problem renders setpoints for gas-lift rates and well head pressures of the wells, denoted by $q^n_{\text{gl,adj}}$ and whp^n_{adj}. The subscript adj refers to the fact that they result from the adjusted values of GOR and BSW.

4.3.2. Control projection

The variables that maximize oil production can be readily applied to the platform, if they are feasible. Since the well models used may be inaccurate, some strategy is necessary to reconcile the theoretical optimal solutions with the real system.

A procedure is proposed by Hanssen and Foss (2015), which suggests the implementation of the solution of daily production optimization problems by departing from the current operating point and making incremental changes until the actual constraints become active. The present work does not focus on how to implement the solution, but on evaluating the potential gains that can be driven by adjusted models. For this reason, we followed a simpler approach which consists in finding feasible values for wellhead pressure and lift-gas flow (for the actual models) that are closest to the ones that maximize oil production (of the adjusted models). This is achieved by solving the following problem

$$\min_{(\mathbf{q}_{\text{gl}},\mathbf{whp})} \sum_{c\in\mathcal{C}} \sum_{n\in\mathcal{N}(c)} \left[(q^n_{\text{gl}} - q^n_{\text{gl,adj}})^2 + (whp^n - whp^n_{\text{adj}})^2\right] \tag{21a}$$

$$\text{s.t. } (5)\text{–}(10) \tag{21b}$$

$$(18),\ (19), \tag{21c}$$

which is a mixed-integer quadratic programming (MIQP) problem, with the objective being the only nonlinear term. Notice that gor^n and bsw^n in (5) refer to the benchmark (error-free) values:

$$gor^n = gor^n_{\text{real}}$$
$$bsw^n = bsw^n_{\text{real}}$$

The final oil production is then computed using the MARLIM simulation models with the operating conditions obtained from the solution of Problem (21), which are feasible for the "real" production system.

5. Case study

Using the methodology presented so far, this section presents the case study used to validate the reconciliation model. To synthesize, the procedures used to test the proposed methodology for parameter identification are illustrated in Fig. 8, the process can be summarized as:

- First, a test scenario is generated and the method of Section 3 is applied, rendering the adjusted values for GOR and BSW of each well;
- Then, the daily production is optimized by solving the MILP problem (20), using the adjusted well models to suggest new operating conditions for the platform according to presented in Section 4;
- This optimal condition is projected onto to the feasible space with the use of error-free models of the platform (with benchmark values of GOR and BSW), by solving MIQP problem (21).
- Finally, the projected operating conditions are introduced in the simulation models, so the final production is computed.

To compare the production obtained with the adjusted parameters with the one induced by historical values (last separator test), the steps above are repeated, except for the identification strategy of Section 3, which is not performed. The error is then defined as the measure of distance between the oil production of adjusted/historical scenarios compared with the benchmark scenarios.

In order to illustrate the above, we start with the data reconciliation of a single scenario. The first step is to solve the problem (16), which renders a vector containing gor^n_{adj} and bsw^n_{adj} for each well $n \in \mathcal{N}$. Fig. 9 compares the values of GOR and

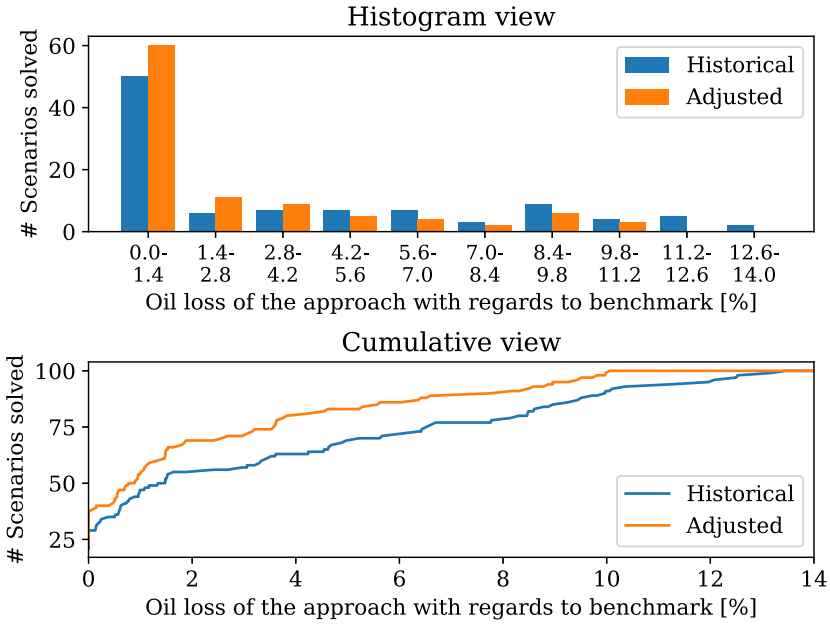


Fig. 14. Number of solutions of each approach, given the loss with respect to the benchmark. Histogram and cumulative view. Case with disturbance of 5% in GOR and BSW, and measurement error of 5% in gas, water and oil production.

Table 1
Final oil production computed for the scenario depicted in Fig. 9.

GOR/BSW parameters	Oil production
Benchmark values	1500.0 m^3/d
Adjusted values	1500.0 m^3/d
Historical values	1316.9 m^3/d

BSW thus adjusted with the historical well-test values (separator test values) and pseudo present-time values for a scenario with 10 wells.

Employing the procedures delineated above, the scenario depicted in Fig. 9 rendered the oil production given in Table 1 for the benchmark (if the operator knew the true parameters), adjusted, and historical parameters. In this example, the proposed method rendered a final oil production almost equal to that of the true parameters.

Next, we present two experiments following the previously presented methodology. In the first, the oil, gas and water measurements are error-free (i.e. the measurement is accurate); in the second experiment, measurement error was added to the reconciliation process.

5.1. Experiment 1: comparison with best of historical and adjusted

In the first experiment, 100 different scenarios were created, where the pseudo present-time values for GOR and BSW of each well varied up to 5% from the historical well-test values. In the second experiment, this variation was up to 10%.

For each scenario, the final oil production obtained by the proposed parameter identification method was compared to that obtained using the historical well test values of GOR and BSW, as illustrated in Fig. 8. Then, the loss of oil in relation to the theoretically best possible (the benchmark) was computed for both approaches. The comparison is presented in Fig. 10. The horizontal axis specifies the loss of oil with regards to the benchmark (error-free) value, with theoretically optimal production, while the vertical axis lists the number of scenario solutions of the approaches. In the histogram (above) we list, for each given range of oil loss, the number of scenario solutions obtained by each approach. The cumulative view (below) accumulates all scenarios solutions with at most the specified loss of oil. While both approaches tend to provide less than 3% of loss, the proposed method has a clear advantage, being the most effective in more than 70% of the cases. It also tends to have errors of smaller magnitude. The errors made by the historical approach tend to be distributed through the whole range considered in the histogram.

Fig. 11 presents the results for the second experiment (where the benchmark values of GOR and BSW were allowed to vary 10%). Though the errors are about twice as large of the previous experiment, the overall behavior is remarkably similar. The proposed methodology is consistently better than using the historical data.

5.1.1. Comparison with theoretical optimum

We also computed the loss incurred for each scenario in relation to the theoretically optimal production. The results are presented in Fig. 12 for the first experiment, and in Fig. 13 for the second. The proposed adjustment method has lower median errors (marked by the colored line). The boxes for the adjusted parameters, which hold 25% of the cases above and below the median, are also smaller, meaning the values are less dispersed. Even the outliers, marked by the circles, have lower magnitude. Outliers in this case represent scenarios in which the reconciliation methodology did not work properly. In general, the results presented show that — given a large number of samples – the method does well in most cases, but can still present poor results in some scenarios.

5.2. Experiment 2: considering measurement error

The experiment here accounts for imprecise measurements for the wells' total oil, gas, and water. The present-time values for GOR and BSW varied 5% from the historical well-test estimates. The flow considered in Eq. (6) had a random error by up to 5%. Again, 100 scenarios were created. The loss of production with regards to the theoretical maximum (benchmark) value is presented

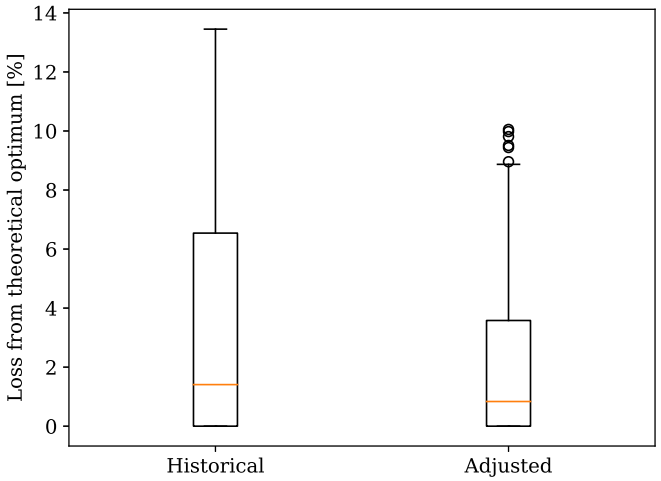


Fig. 15. Distribution of oil loss with respect to the theoretical optimum, for both approaches. Case with disturbance of 5% in GOR and BSW, and measurement error of 5% in gas, water and oil production.

in Fig. 14, and the distribution of the losses is given in the box plot of Fig. 15. The errors in oil flow may affect the overall gains of the proposed methodology, but the use of adjusted parameters still has a clear advantage with regards to using historical data nonetheless. In all of the numerical experiments, the proposed method for adjusting the GOR and BSW leads to a production of oil with lower median loss, and the losses are less dispersed.

6. Conclusion

This work presented a methodology for automatically detecting steady-state and re-estimating GOR and BSW of individual wells that produce to a common separator, accounting for platform constraints. Dedicated procedures have been successfully employed to detect when wells are in a stationary state automatically.

In the numerical experiments, we showed that the adjusted values for GOR and BSW enable a more efficient maximization of oil production, compared to the well test values. The results were consistent, in that the proposed approach had smaller and fewer losses of oil compared to the approach that used past well test values.

With the methodology proposed in the paper, GOR and BSW values are updated to reflect the prevailing operating conditions. Given the continuous measurement of the platform flow values (as a whole), the proposed methodology can aid operators in finding GOR and BSW values closer to the actual values at any time a stationary regime is identified. In other words, it is no longer necessary to wait for a new separation test to update the given well parameters, but instead, wait for the next steady-state regime.

One limitation of this method is its dependence of steady-state information of the wells, which can increase the time-frame of the updates. To earlier identify well productions, dynamic models would be required. This is performed in the work of Krishnamoorthy et al. (2018) for a case in which two wells are producing to a common manifold.

Future developments of this study will include more complex systems, and consider other tuning parameters, such as the *Productivity Index*. It is also possible to reconcile with the measurements of more variables (pressures in flow-lines, for example), to ensure the adjusted model stay close to the current situation. Finally, Petrobras aims to refine this methodology and evaluate it in field tests in a real platform.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Laio Oriel Seman: Software, Investigation, Writing - original draft, Conceptualization. **Caio Merlini Giuliani:** Investigation, Writing - original draft, Conceptualization. **Eduardo Camponogara:** Investigation, Project administration, Writing - original draft, Conceptualization. **Eduardo Rauh Müller:** Software, Investigation, Writing - review & editing, Conceptualization. **Bruno Ferreira Vieira:** Validation, Resources, Conceptualization. **Luis Kin Miyatake:** Validation, Resources, Conceptualization. **André Gonçalves Medeiros:** Validation, Resources, Conceptualization.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.compchemeng.2020.107179.

References

Beale, E.M.L., Tomlin, J.A., 1970. Special facilities in a general mathematical programming system for non-convex problems using ordered sets of variables. In: Lawrence, J. (Ed.), Proceedings of the Fifth International Conference on Operational Research, p. 447454.

Belotti, P., Lee, J., Liberti, L., Margot, F., Wächter, A., 2009. Branching and bounds tightening techniques for non-convex MINLP. Optim. Methods Softw. 24 (4-5), 597–634. doi:10.1080/10556780903087124.

Bikmukhametov, T., Jäschke, J., 2020. Combining machine learning and process engineering physics towards enhanced accuracy and explainability of data-driven models. Comput. Chem. Eng. 138, 106834. doi:10.1016/j.compchemeng.2020.106834.

Câmara, M., Quelhas, A., Pinto, J., 2016. Performance evaluation of real industrial RTO systems. Processes 4 (4), 44. doi:10.3390/pr4040044.

Chachuat, B., Srinivasan, B., Bonvin, D., 2009. Adaptation strategies for real-time optimization. Comput. Chem. Eng. 33 (10), 1557–1567. doi:10.1016/j.compchemeng.2009.04.014.

Chen, C.Y., Joseph, B., 1987. On-line optimization using a two-phase approach: an application study. Ind. Eng. Chem. Res. 26 (9), 1924–1930. doi:10.1021/ie00069a034.

Floudas, C.A., 1995. Nonlinear and Mixed-Integer Optimization: Fundamentals and Applications, 1st edition Oxford University Press.

Foss, B., Knudsen, B.R., Grimstad, B., 2018. Petroleum production optimization – a static or dynamic problem? Comput. Chem. Eng. 114, 245–253. doi:10.1016/j.compchemeng.2017.10.009.

Foss, B.A., Jensen, J.P., 2011. Performance analysis for closed-loop reservoir management. SPE J. 16 (01), 183–190. doi:10.2118/138891-pa.

Golub, G.H., Loan, C.F.V., 1996. Matrix Computations, 3 Baltimore Johns Hopkins University Press.

Grimstad, B., Foss, B., Heddle, R., Woodman, M., 2016. Global optimization of multiphase flow networks using spline surrogate models. Comput. Chem. Eng. 84, 237–254. doi:10.1016/j.compchemeng.2015.08.022.

Gunnerud, V., Foss, B., 2010. Oil production optimization—a piecewise linear model, solved with two decomposition strategies. Comput. Chem. Eng. 34 (11), 1803–1812. doi:10.1016/j.compchemeng.2009.10.019.

Gurobi Optimization, L., 2020. Gurobi optimizer reference manual.

Hanssen, K.G., Foss, B., 2015. Production optimization under uncertainty - applied to petroleum production. IFAC-PapersOnLine 48, 217–222. doi:10.1016/j.ifacol.2015.08.184.

Hoffmann, A., Stanko, M., 2017. Short-term model-based production optimization of a surface production network with electric submersible pumps using piecewise-linear functions. J. Pet. Sci. Eng. 158, 570–584. doi:10.1016/j.petrol.2017.08.063.

Hülse, E.O., Silva, T.L., Camponogara, E., Rosa, V.R., Vieira, B.F., Teixeira, A.F., 2020. Introducing approximate well dynamics into production optimization for operations scheduling. Comput. Chem. Eng. 136, 106773. doi:10.1016/j.compchemeng.2020.106773.

IBM ILOG CPLEX, 2009. V12. 1: users manual for CPLEX. Int. Bus. Mach. Corp. 46 (53), 157.
Jang, S.-S., Joseph, B., Mukai, H., 1987. On-line optimization of constrained multivariable chemical processes. AIChE J. 33 (1), 26–35. doi:10.1002/aic.690330105.
Kim, S.J., Koh, K., Boyd, S., Gorinevsky, D., 2009. l_1 trend filtering. SIAM Rev. 51 (2), 339–360. doi:10.1137/070690274.
Krishnamoorthy, D., Foss, B., Skogestad, S., 2018. Steady-state real-time optimization using transient measurements. Comput. Chem. Eng. 115, 34–45. doi:10.1016/j.compchemeng.2018.03.021.
McCormick, G.P., 1976. Computability of global solutions to factorable nonconvex programs: Part i — convex underestimating problems. Math. Program. 10 (1), 147–175.
Merlini Giuliani, C., Camponogara, E., 2015. Derivative-free methods applied to daily production optimization of gas-lifted oil fields. Comput. Chem. Eng. 75, 60–64. doi:10.1016/j.compchemeng.2015.01.014.
Rincón, F., Roux, G., Lima, F., 2015. A novel ARX-based approach for the steady-state identification analysis of industrial depropanizer column datasets. Processes 3 (2), 257–285. doi:10.3390/pr3020257.
Sahinidis, N.V., 1996. BARON: a general purpose global optimization software package. J. Global Optim. 8 (2), 201–205. doi:10.1007/BF00138693.
Schlumberger, 2012. Olga dynamic multiphase flow simulator. V 7.1.3.
Schlumberger, A., 2009. Pipesim user manual, well design. Drilling and Petroleum.
Seman, L.O., Miyatake, L.K., Camponogara, E., Giuliani, C.M., Vieira, B.F., 2020. Derivative-free parameter tuning for a well multiphase flow simulator. J. Pet. Sci. Eng. 107288. doi:10.1016/j.petrol.2020.107288.
Silva, T.L., Camponogara, E., 2014. A computational analysis of multidimensional piecewise-linear models with applications to oil production optimization. Eur. J. Oper. Res. 232 (3), 630–642. doi:10.1016/j.ejor.2013.07.040.
Tao, L., Li, C., Kong, X., Qian, F., 2012. Steady-state identification with gross errors for industrial process units. In: Proceedings of the 10th IEEE World Congress on Intelligent Control and Automation doi:10.1109/wcica.2012.6359172.
Vielma, J.P., Ahmed, S., Nemhauser, G., 2010. Mixed-integer models for nonseparable piecewise-linear optimization: unifying framework and extensions. Oper. Res. 58 (2), 303–315. doi:10.1287/opre.1090.0721.
Wächter, A., Biegler, L.T., 2006. On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming. Math. Program. 106 (1), 25–57. doi:10.1007/s10107-004-0559-y.